
Uncovering Conceptual Blindspots in Generative Image Models Using Sparse Autoencoders

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Abstract

Despite their strong empirical performance, generative image models trained on large-scale datasets systematically fail to synthesize certain seemingly elementary concepts—e.g., human hands or objects appearing in groups of four—whose occurrence in the training distribution is a priori expected. To date, such failure modes have been characterized only anecdotally, leaving unresolved whether they constitute isolated, concept-specific anomalies or reflect a structural limitation of the learned distribution. We address this question by formalizing and operationalizing the notion of a “conceptual blindspot”—a concept whose prevalence in the training distribution is not matched by its prevalence in the model’s output distribution, yielding either omission or misrepresentation. The proposed method employs sparse autoencoders (SAEs) to recover an interpretable, low-dimensional concept basis, under which concept prevalence can be quantitatively compared between the natural (real) and generated image distributions. Concretely, we train an archetypal SAE (RA-SAE) on DINOv2 features with a dictionary of 32,000 concepts—to our knowledge the largest such SAE reported—thereby enabling fine-grained quantification of per-concept distributional disparities. Applied to four widely used generative models (Stable Diffusion 1.5/2.1, PixArt, and Kandinsky), the method identifies concrete suppressed blindspots (e.g., bird feeders, DVD discs, and whitespaces on documents) and exaggerated blindspots (e.g., wood background texture and palm trees). At the level of individual datapoints, the analysis further isolates memorization artifacts, defined as instances in which a model reproduces highly specific visual templates observed during training. Overall, we contribute a theoretically grounded framework that systematically identifies conceptual blindspots by quantifying a generative model’s conceptual fidelity relative to the underlying data-generating process.

1 Introduction

Generative image models trained on large-scale datasets have attained unprecedented capabilities, supporting applications both within the vision domain [1–9] and well beyond it [9–15]. This progress notwithstanding, both qualitative [16–18] and quantitative [19–20] studies report that such models can fail to synthesize comparatively elementary concepts, including human hands [21–24], objects appearing in groups of four [25], and negations or object relations [20, 26]. More precisely, when conditioned on prompts specifying such concepts, these models tend to generate outputs that are structurally related to, but not coincident with, the target concept—for instance, rendering hands with six fingers. We refer to these failure modes as “conceptual blindspots”¹. Such

¹We borrow the term “blindspots” from psychology literature [27], wherein it is used to refer to scenarios where an agent makes biased decisions despite exposure to observations that contradicts their decision’s rationale.



Figure 1: The energy-difference statistic $\delta(k)$ quantifies the extent to which a model over- or under-generates concept c_k relative to its empirical frequency in natural data. We classify concepts with $\delta(k) < 0.1$ as **suppressed conceptual blindspots** and concepts with $\delta(k) > 0.9$ as **exaggerated conceptual blindspots**. The displayed images, produced by four widely used generative image models, illustrate instances of both conceptual blindspots and concepts that are faithfully reproduced (aligned).

behavior is counterintuitive, given that the training corpus can reasonably be assumed to contain a substantial number of demonstrations that accurately depict the concepts in question. This motivates a central question: do these failures constitute idiosyncratic quirks confined to specific concepts, or are they instances of a systematic phenomenon in which, across a broad range of concepts, a model is either unable to generate the concept or is disproportionately likely to do so?

This question cannot be resolved with existing protocols for evaluating generative image models [28, 29]. Prevailing protocols rely on coarse, image-level statistics designed to assess sample realism, e.g., FID [30]; such statistics aggregate over the full image and therefore do not resolve distributional failures localized to specific concepts. Diversity- and coverage-oriented statistics, such as those derived from CLIPScore [31–33], provide only partial evidence, as they operate at the level of whole images or prompts rather than at the granularity of individual features or concepts [34, 35], and thus cannot localize conceptual blindspots. Finally, qualitative evaluation protocols—e.g., participant surveys [5, 17, 36–38] or open-ended exploration [39]—can surface concept-level failures, but do not constitute a scalable, reproducible methodology that generalizes across model families and admits systematic cross-model comparison.

This work. Motivated by the above, we contend that identifying and analyzing conceptual blindspots requires a methodology that, in an automated and unsupervised fashion, isolates concepts in the data distribution whose probability of generation under the true data-generating process differs from that under the trained model. Accordingly, this work makes three contributions:

- **Formalizing conceptual blindspots in generative image models.** We introduce a formal framework for defining and quantifying conceptual blindspots relative to the natural image distribution (Section 3). By grounding the definition in the data-generating process rather than in anecdotal or human-specified concept sets, the framework yields a principled, quantitative characterization of model limitations.
- **A scalable, unsupervised procedure for identifying conceptual blindspots via sparse autoencoders.** We instantiate this framework as a structured pipeline that uses sparse autoencoders (SAEs) to extract and compare per-concept prevalence between natural and synthesized images (Section 4). To this end, we train and release an archetypal SAE (RA-SAE) on DINOv2 features with a dictionary of 32,000 concepts—to our knowledge the largest RA-SAE reported.
- **Exploratory tooling and empirical analysis.** We provide an interactive web tool supporting both distribution- and datapoint-level analysis of blindspots across models (Sections 5.1–5.4), and apply the method to Kandinsky, PixArt, and Stable Diffusion (SD) 1.5/2.1 (Section 5). The analysis surfaces concrete instances of suppressed conceptual blindspots (e.g., bird feeders, DVD discs, and whitespaces on documents) and exaggerated conceptual blindspots (e.g., wood background texture and palm trees), as shown in Fig. 1.

The trained model, source code, and web tool are released in a public software repository.

72 2 Related Work

73 2.1 Explainability in Vision

74 Early work in explainable AI for computer vision centered on attributing model outputs to influ-
75 ential input regions [40–42]. Such attribution methods convey limited semantic information about
76 the learned representation and have been shown to yield unreliable explanations under controlled
77 sanity checks [43–45]. To overcome these limitations, concept-based interpretability [46] was devel-
78 oped to identify semantically meaningful directions in the representation space of neural networks,
79 characterizing not only *where* a model attends but *which* concepts and structures it encodes [47–49].

80 Recent analysis establishes that widely used concept-based interpretability methods—ACE [44],
81 CRAFT [48], and Sparse Autoencoders (SAEs) [50–51]—solve a common dictionary-learning
82 problem under differing constraints [52]. Among these, SAEs have proven particularly scalable
83 for concept-based interpretability. Although the original SAE formulation exhibits documented
84 limitations—including excessively specific features [53], difficulties with compositionality [54], and
85 limited causal effect under intervention [55]—subsequent variants have addressed several of these
86 issues, notably the archetypal SAE (RA-SAE) [56].

87 Complementary interpretability techniques include prompt-based probing [57], visualization of
88 attention maps or activations [58–59], and dataset-level statistics [32–33] such as diversity or
89 distribution-coverage metrics; each affords only partial evidence toward the questions we pose.
90 Critically, these techniques operate at the granularity of whole images or prompts rather than of
91 individual features or concepts [34–35]. They further rely on subjective interpretation and do not
92 disambiguate distinct failure modes [60]. Consequently, existing methods and metrics are insufficient
93 for systematically localizing feature- and concept-level deficiencies of generative image models [28].

94 2.2 Generative Image Models

95 Diffusion-based methods are now the dominant paradigm across modalities in generative vision
96 modeling, spanning image [4–61–64], video [65–72], and 3D [7–73–75] generation. In the image
97 domain, the paradigm originates with denoising diffusion probabilistic models (DDPMs) [76],
98 subsequently generalized to non-Markovian diffusion processes by denoising diffusion implicit
99 models (DDIMs) [77].

100 The Stable Diffusion (SD) [9] model family made DDPMs broadly accessible to both research and
101 open-source communities. The original release was succeeded by several versions, including SD
102 2 [78], SD 3 [79], and SD XL [80]. Numerous architectural extensions impose additional conditioning
103 on the diffusion process—e.g., style [81–82], pose [83], and identity [84–85]—and admit alternative
104 input modalities such as image-to-image generation. A range of latent diffusion models built upon
105 SD—including Kandinsky [86], PixArt [87], DeepFloyd [4–88], and FLUX [89]—have likewise
106 been proposed.

107 2.3 Datasets for Generative Image Models

108 The recent success of generative vision models is attributable in large part to the availability of
109 substantial compute and large-scale web-crawled datasets [90–92]. In particular, LAION-5B [93] has
110 been central to the training of open-source text-to-image models, including SD and its derivatives.
111 Constructed from Common Crawl [94], this dataset comprises over 5B image-caption pairs, of which
112 2.3B are English. Other prominent corpora include COYO-700M [95] and Conceptual Captions [96],
113 containing 700M and 3M image-caption pairs, respectively.

114 As LAION-5B was more widely adopted, documented concerns regarding its biases accumulated [97–
115 101]. Filtering notwithstanding, harmful content persisted [97–99], including NSFW material [100]
116 and hundreds of CSAM instances [101], which prompted its temporary withdrawal from official
117 distribution channels. The dataset additionally exhibits low-quality images [102] and web-style
118 captions (e.g., product descriptions) that are distributionally misaligned with the prompts users issue
119 to trained models [103].

2.4 Concept Discovery and Sparse Coding in Generative Image Models

Dictionary learning seeks sparse representations of input data in which each sample is reconstructed as a linear combination of a small number of dictionary atoms [104, 106]. Grounded in compressed-sensing theory [107, 108], the field has progressed from early vector-quantization methods [109] to more expressive formulations including Non-negative Matrix Factorization [110, 111], Sparse PCA [112], and K-SVD [113]. More recent developments encompass online estimation [114], structured sparsity [115], and identifiability guarantees [116, 117], together with increasingly explicit connections to deep learning [118, 119].

Sparse-coding techniques have also been applied to characterize the emergence of high-level concepts within diffusion models [120]. Prior to diffusion models, concept-grounded interpretability was applied to earlier generative architectures via concept-bottleneck models, which require human supervision at training time [121], and via post-hoc detectors that retrofit concept supervision [122]. Both approaches presuppose a human-specified concept vocabulary and, by construction, cannot capture broader trends over concepts the user has not enumerated in advance.

Aggregate quality metrics—e.g., precision, recall, density, and coverage [35, 123]—and latent density scores, which predict sample quality from the model’s latent space [124], have been proposed to evaluate generative image models. Although effective at exposing distribution-level gaps, these metrics provide little insight into which specific concepts are under- or over-represented.

Collectively, these limitations motivate a scalable, unsupervised framework capable of systematically localizing and quantifying concept-level failure modes in generative image models [125].

3 Formalizing Conceptual Blindspots in Generative Models

We first formalize *conceptual blindspots* as systematic discrepancies between the conceptual content of natural images and that of model-generated outputs. This formalization furnishes principled, quantitative measures that identify which concepts a generative model under- or over-represents relative to its data distribution. The overall procedure is depicted in Fig. 2. The derivation invokes a set of standard assumptions from the identifiability literature [126–130]; empirically, however, we observe that meaningful phenomenology is elicited even when these assumptions are only approximately satisfied.

Definition 1 (Data-Generating Process). Let $\mathcal{C} \subset \mathbb{R}^K$ denote a latent space with a Boltzmann prior $p(c) = \exp(-E(c))Z^{-1}$, where $E(\cdot)$ denotes an energy function that linearly decomposes over individual latents and Z is the corresponding partition function, i.e., $E(c) = \sum_k E(c_k)$ and hence $p(c) = \prod_k p_k(c_k)$. A data-generating process (DGP) is an invertible function $G : \mathcal{C} \rightarrow \mathcal{X}$ that maps the latents $c \in \mathcal{C}$ to observations $x \in \mathcal{X}$, i.e., $x = G(c)$.

For notational economy, we let $p(\cdot)$ denote both the latent density $p(c)$ and its pushforward to image space $p(x)$, with $x = G(c)$. This overloading is well-defined because the invertibility of G induces, via the change-of-variables formula, a valid distribution over $\mathcal{X}$. Concretely, the individual coordinates of the latent space correspond to the **concepts** underlying the data distribution $P_{\mathcal{X}}$, itself defined over an observation space of images $\mathcal{X}$. For instance, distinct latent coordinates may encode concepts such as color, shape, size, and location [131, 132]. We additionally assume the data-generating process associates a text description $t \in \mathcal{T}$ with each sampled image, though we do not model this association explicitly. Such text descriptions serve as conditioning inputs for a text-conditioned **generative image model** g_{θ} with parameters θ , trained on image-text pairs to map a noise sample $\eta \sim \mathcal{N}(0, I)$ together with a scene description t to an image x' realizing that description.

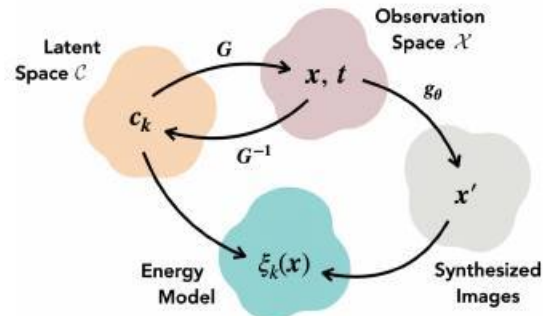


Figure 2: **Schematic of the proposed framework.** Latent concepts $c_k \in \mathcal{C}$ are mapped to observations (x, t) by the invertible data-generating process (DGP). The generative model S_{θ} synthesizes images x' conditioned on t . The energy model $\xi_k(x)$ recovers per-concept representations from both x and x' , enabling a distributional comparison from which blindspots are identified.

To define conceptual blindspots of the model g_θ , we compare the probability mass a concept receives under the data-generating process with the mass it receives under the model. To this end, we take an evaluation set of natural image-text pairs $(\mathcal{D}_\mathcal{X}, \mathcal{D}_\mathcal{T})$ and construct a corresponding set of synthesized images $\mathcal{D}'_\mathcal{X}$ by conditioning g_θ on the associated text descriptions. To estimate the probability of occurrence of a concept, we introduce an **energy model** $\xi : \mathcal{X} \rightarrow \mathbb{R}^d$ that maps an image to a d -dimensional vector whose k^{th} coordinate is a scalar quantifying the energy of the k^{th} concept, i.e., $\xi_k(\mathbf{x}) = E(c_k)$. These estimates are well-posed under the assumed invertibility of the data-generating process. The probability the data-generating process assigns to image $\mathbf{x}$ is then $p_k(\mathbf{x}) \propto \exp(-\sum_k \xi_k(\mathbf{x}))$, with partition function Z_ξ , which induces the population-level estimate $p_k(\mathcal{D}_\mathcal{X}) = \prod_{\mathbf{x} \in \mathcal{D}_\mathcal{X}} p_k(\mathbf{x})$. Combining this construction with the logistic sigmoid map $\sigma(\cdot)$, we define below the average energy difference for the k^{th} concept between the datasets $\mathcal{D}_\mathcal{X}$ and $\mathcal{D}'_\mathcal{X}$.

Definition 2 (Energy Difference). Let $\mathbf{x} \in \mathcal{D}_\mathcal{X}$ denote a real image sampled from the data-generating process G , and let $\mathbf{x}' \in \mathcal{D}'_\mathcal{X}$ be a synthetic image generated by the model g_θ . Let $\xi_k : \mathcal{X} \rightarrow \mathbb{R}$ denote the energy assigned to the k^{th} concept by the energy model ξ . We define the energy difference for concept k as:

$$\begin{aligned} \delta_{g_\theta \leftrightarrow G}(k) &= \sigma(\mathbb{E}_{\mathbf{x}'}[\xi_k(\mathbf{x}')] - \mathbb{E}_{\mathbf{x}}[\xi_k(\mathbf{x})]) \\ &= \frac{p_k(\mathcal{D}'_\mathcal{X})}{p_k(\mathcal{D}_\mathcal{X}) + p_k(\mathcal{D}'_\mathcal{X})}, \end{aligned} \quad (1)$$

where the expectations are taken over $\mathcal{D}'_\mathcal{X}$ and $\mathcal{D}_\mathcal{X}$, respectively, and $p_k(\mathcal{D}) \propto \exp(-\sum_{\mathbf{x} \in \mathcal{D}} \xi_k(\mathbf{x}))$ denotes the unnormalized conceptual probability mass of dataset $\mathcal{D}$ under concept k .

Thus, the energy difference for the k^{th} concept is the normalized odds that the concept occurs in a target set of observations (here, $\mathcal{D}'_\mathcal{X}$) relative to a baseline set (here, $\mathcal{D}_\mathcal{X}$). On the basis of this measure, we now define conceptual blindspots.

Definition 3 (Suppressed / Exaggerated Conceptual Blindspots). Given a generative image model g_θ , we say, compared to the data-generating process G , c_k is a suppressed conceptual blindspot in the model if $\delta_{g_\theta \leftrightarrow G}(k) < \lambda_{\min}$ and exaggerated if $\delta_{g_\theta \leftrightarrow G}(k) > \lambda_{\max}$.

In summary, a conceptual blindspot is a concept whose probability of occurrence in generated images departs markedly—through either suppression or exaggeration—from its prevalence under the data-generating process. Suppressed concepts exhibit disproportionately low activation ($\delta(k) < \lambda_{\min}$), whereas exaggerated concepts are over-represented ($\delta(k) > \lambda_{\max}$). Throughout the analysis we fix the thresholds $\lambda_{\min} = 0.1$ and $\lambda_{\max} = 0.9$ to delimit these two regimes.

This definition is related to the notion of “mode collapse” studied in prior work (e.g., Bau et al. [39]), the distinction being the granularity of the analysis. Whereas mode collapse concerns the elevated or suppressed odds of generating *entire images*, our formulation targets shifts in the odds of individual concepts. For example, if a model systematically fails to generate objects with a white background, we classify that concept as a suppressed conceptual blindspot.

4 Method: Operationalizing the Definition of Conceptual Blindspots

We now describe the pipeline that operationalizes the definitions of Sec. 3 for a given generative model g_θ . Two objects are required: (i) a set of images sampled from g_θ that can be placed in correspondence with the ground-truth generative process, and (ii) an energy model that renders this correspondence quantitatively comparable. Notation. We write $\|\cdot\|_F$ for the Frobenius norm and $\|\cdot\|_0$ for the number of non-zero entries (the ℓ_0 pseudo-norm). For a vector or matrix $\mathbf{X}$, the inequality $\mathbf{X} \geq 0$ is understood element-wise. For $n > 0$ we set $[n] := \{1, \dots, n\}$, and denote the i -th row of a matrix $\mathbf{A}$ by $\mathbf{A}_i$.

From Prompts to Latent Representations. To localize conceptual blindspots in a model g_θ , we compare a dataset $\mathcal{D}_\mathcal{X}$ of image-caption pairs $(\mathbf{x}, t)$ drawn from the data-generating process G against their synthetic counterparts, obtained by conditioning g_θ on the same text descriptions.

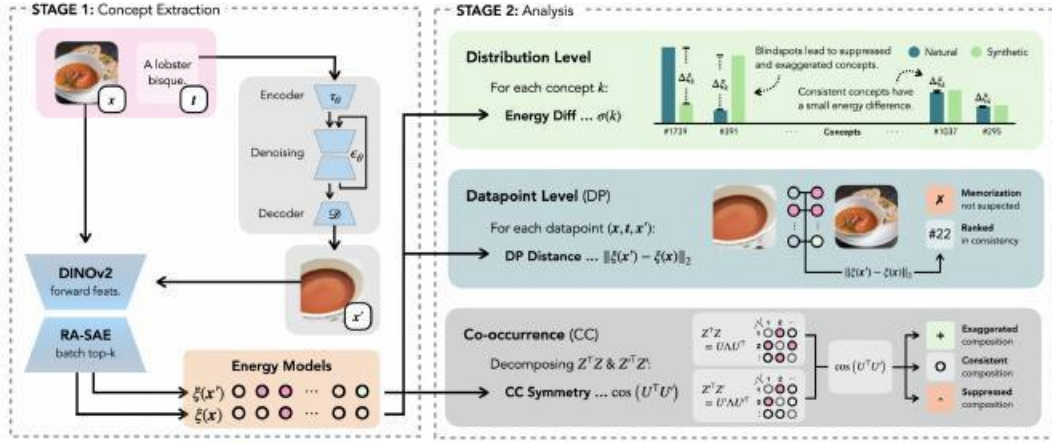


Figure 3: **Concept extraction pipeline.** For each triplet (x, t, x') , the DINOv2 features of x and x' are computed and mapped by an RA-SAE into sparse embeddings, yielding the energy representations $\xi(x)$ and $\xi(x')$, respectively. The k^{th} coordinate $\xi_k(x) = E(c_k)$ holds the energy of the k^{th} concept. Formally, given a caption t , we synthesize a counterpart image x' with a pretrained text-to-image generator $g_\theta: \mathcal{T} \rightarrow \mathcal{X}$, instantiated as a denoising diffusion probabilistic model (DDPM) [78][86][87]. Generation proceeds in latent space along the reverse diffusion trajectory $(\gamma_t)_{t=0}^T$:

$$\gamma_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad \gamma_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\gamma_t - \frac{1-\alpha_t}{\sqrt{1-\alpha_t}} \epsilon_\theta(\gamma_t, t, c) \right) + \sigma_t \eta_t, \quad \text{and} \quad \eta_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}),$$

where $\alpha_t \in (0, 1)$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$ are prescribed by the standard cosine noise schedule. The terminal latent γ_0 is decoded by a pretrained VAE to produce the synthetic image $x' = \text{VAE}(\gamma_0)$. For the remainder of the paper we abstract g_θ as a black-box map from prompts to images, $t \mapsto x'$.

Defining the Energy Model. Leveraging prior results establishing that self-supervised objectives can invert the data-generating process and recover its underlying energy function up to a linear transformation [126][128][133][134], we adopt DINOv2 [135], a state-of-the-art self-supervised encoder, as our feature extractor. Under the regime in which the number of concepts underlying the DGP exceeds the dimensionality of the encoder’s feature space [51][136], we fit sparse autoencoders (SAEs) to these features to recover the subspaces corresponding to individual concepts [50][56][137][139]. The underlying rationale is that if DINOv2 encodes the generative concepts along approximately orthogonal directions (consistent with the independence assumption in Def. 1), then an SAE can disentangle these concepts onto individual coordinates of its latent space [136]. The activation the SAE assigns to a given coordinate then serves as our estimate of the ground-truth energy of the concept encoded by that coordinate.

Formally, let $f: \mathcal{X} \rightarrow \mathbb{R}^d$ denote the feature extractor (the DINOv2 encoder); we compute features $\mathbf{a} = f(x) \in \mathbb{R}^d$ for both natural and synthetic images from $\mathcal{D}_\mathcal{X}$ and $\mathcal{D}'_\mathcal{X}$. Denoting by n the number of image-text pairs, we stack the real and generated feature vectors into matrices $\mathbf{A}, \mathbf{A}' \in \mathbb{R}^{n \times d}$. Each feature vector is then decomposed into a sparse combination of learned concept atoms via an SAE. Specifically, let $\mathbf{D} \in \mathbb{R}^{d \times K'}$ be a dictionary of K' concept vectors and let $\Psi: \mathbb{R}^d \rightarrow \mathbb{R}^{K'}$ denote the SAE encoder mapping features to sparse codes. Applying Ψ row-wise produces the activation matrix $\mathbf{Z} = \Psi(\mathbf{A}) \in \mathbb{R}^{n \times K'}$, whose i -th row $\mathbf{z}_i = \Psi(\mathbf{a}_i)$ is the sparse concept decomposition of image i . The decoder reconstructs features as $\mathbf{Z}\mathbf{D}^\top$, and the SAE is trained to minimize reconstruction error subject to sparsity and non-negativity constraints:

$$\min_{\Psi, \mathbf{D}} \|\mathbf{A} - \Psi(\mathbf{A})\mathbf{D}^\top\|_F^2 \quad \text{s.t.} \quad \Psi(\mathbf{A}) \geq 0, \|\Psi(\mathbf{A})_i\|_0 \ll K' \quad \forall i \in [n]. \quad (2)$$

Vanilla SAEs tend to converge to dictionaries with arbitrary orientation, rendering downstream analyses highly sensitive to the random seed. To eliminate this instability and ensure seed-independent, reproducible results, we adopt the *Archetypal* SAE (RA-SAE) [56] together with a TOP-K sparsity constraint [137]. RA-SAE restricts the dictionary $\mathbf{D}$ to lie in the convex hull of the training data. Concretely, we parameterize $\mathbf{D} = \mathbf{W}\mathbf{A}$ with $\mathbf{W} \in \Omega_{K', n}$, where $\Omega_{K', n}$ is the set of row-stochastic matrices in $\mathbb{R}^{K' \times n}$:

$$\Omega_{K', n} := \left\{ \mathbf{W} \in \mathbb{R}^{K' \times n} \mid \mathbf{W} \geq 0, \mathbf{W}\mathbf{1} = \mathbf{1} \right\}. \quad (3)$$

Consequently, every atom $\mathbf{D}_i$ lies in the convex hull of the data $\text{conv}(\mathbf{A})$, and any reconstruction $\mathbf{Z}\mathbf{D}^\top$ lies in the conic hull $\text{cone}(\mathbf{A})$. This constraint guarantees that the learned concepts remain supported on the data distribution [56]. Once trained, the SAE yields a consistent set of sparse codes: $\mathbf{Z}$ for real images and $\mathbf{Z}'$ for their generated counterparts. *These codes represent the same prompt-conditioned visual semantics over a shared, interpretable concept basis, and the per-concept activation values serve as the energy estimates used in our blindspot analysis.* In summary, the method defines a structured pipeline that, for a prompt and its associated real image (t, x) , returns two sparse concept vectors (z, z') , permitting a direct comparison of real and generated visual content within a common conceptual basis.

The complete pipeline is thus as follows: starting from a captioned image (t, x) , we synthesize a counterpart x' and embed both images into a shared sparse concept space via the vision encoder and the trained SAE, obtaining interpretable representations (z, z') that constitute the basis for quantifying the conceptual shifts induced by the generative process.

5 Results

We evaluate four generative image models trained on LAION-5B—SD 1.5, SD 2.1, PixArt, and Kandinsky—on $|\mathcal{X}| = 10,000$ image-text pairs together with their corresponding generations (Fig. 4). The evaluation is organized along three levels of granularity (Fig. 3): a **distribution-level** analysis, which identifies suppressed and exaggerated concepts; a **datapoint-level** analysis, which isolates failures attributable to ambiguity, omission, and memorization; and a **compositional** analysis, which exposes fine-grained distortions in the geometry of concept co-occurrence.

The core contribution at this stage is an interactive exploratory tool, illustrated in Fig. 5. It is a web-based interface organized around a UMAP projection of concept representations, supporting visualization and comparison of concept-level energy differences. The tool is released publicly in a software repository, together with precomputed energy-difference data for the four evaluated models (SD 1.5, SD 2.1, PixArt, and Kandinsky). All subsequent analyses in this paper derive from observations enabled by this tool. Its principal functionalities are:

- **Cross-model and cross-architecture comparison.** For each evaluated model, the tool renders a UMAP embedding of all 32,000 RA-SAE concepts; each point denotes an individual concept, color-coded by its energy difference.
- **Concept-level inspection.** Each concept is associated with a card reporting summary statistics, representative real and generated images (x, x') , and its co-occurrence pattern.
- **Blindspot exploration.** Beyond the UMAP and per-concept views, the tool provides global rankings of suppressed and exaggerated blindspots, surfacing the most pronounced conceptual blindspots.

Rather than exhaustively examining any single phenomenon, we report high-level findings that demonstrate the versatility of the tool and enable broader, user-directed exploration.

5.1 Generative Image Models Cannot Produce Unseen Concepts

To quantify the disparity between each generative model and the natural image distribution, we first analyze the marginal energy difference $\delta(k)$ over the 32,000 concepts recovered by the RA-SAE. As defined in Sec. 4, this statistic measures the relative prevalence of each concept in the synthesized versus natural image sets: $\delta(k) < 0.1$ indicates that concept k is under-represented (suppressed) in the generated images, whereas $\delta(k) > 0.9$ indicates over-representation (exaggeration). Fig. 6 reports the empirical distribution of $\delta(k)$ for each of the four evaluated models. All models exhibit heavy-tailed histograms with substantial mass at both extremes, consistent with systematic discrepancies in concept coverage. Crucially, the left tail—corresponding to suppressed concepts—is denser and longer than the right, indicating a consistent bias toward concept suppression. This asymmetry is



Figure 4: **Representative image-caption pairs** sampled from LAION-5B, paired with images synthesized from the identical prompt by SD 1.5, SD 2.1, PixArt, and Kandinsky.

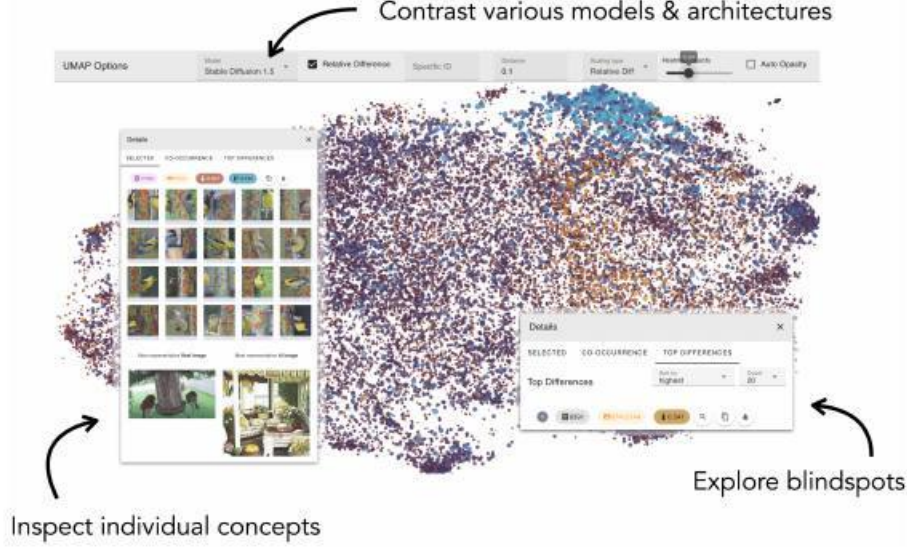


Figure 5: **Overview of the exploratory tool.** The web interface renders a UMAP projection for each evaluated model, in which each point denotes a concept color-coded by its energy difference. Selecting a concept opens a detail panel reporting illustrative images, summary statistics, and the most representative natural and generated images x and x' . An ordered list of the concept’s co-occurrences is displayed alongside global blindspot rankings.

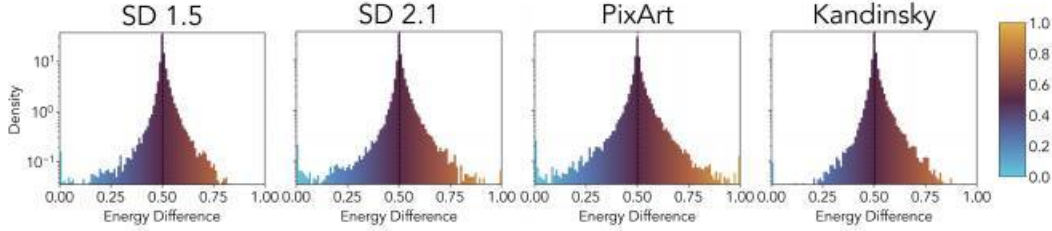


Figure 6: **Concept energy distribution.** Log-scale histograms of the energy difference $\delta(k)$ over all 32,000 concepts, comparing the natural and synthesized distributions for each evaluated model. Mass to the left of zero corresponds to suppressed (under-represented) concepts; mass to the right corresponds to exaggerated (over-represented) concepts.

quantified by the negative sample skewness of each distribution: -0.54 for SD 2.1, -0.40 for both SD 1.5 and PixArt, and -0.23 for Kandinsky.

Although all models display both suppressed and exaggerated concepts, their per-model profiles differ. PixArt, for instance, exhibits a wider spread, indicative of a more heavily suppressed concept distribution. Nonetheless, the consistent left-skew across all four distributions points to a shared tendency toward concept omission; the finer structure of this behavior warrants further analysis, which we pursue in the following sections.

5.2 Structure and Specificity of Conceptual Blindspots

Whereas the previous section quantified marginal discrepancies in concept frequency, we now examine their global geometric structure by embedding all 32,000 concepts into two dimensions with UMAP applied to the sparse codes, coloring each concept by its $\delta(\cdot)$ value. As Fig. 7 shows, well-separated concept clusters emerge across all models. These clusters frequently coincide with contiguous regions of conceptual blindspots, most prominently for suppressed (blue) concepts, indicating that blindspots are spatially structured rather than random—consistent with shared biases in the training distribution or in architectural priors. To assess the cross-model consistency of these patterns, we further analyze both the magnitude and the structure of the concept-level $\delta(\cdot)$ vectors. Fig. 8 reports pairwise scatter plots and Pearson correlation coefficients between the $\delta(k)$ vector of SD 1.5 and those of the

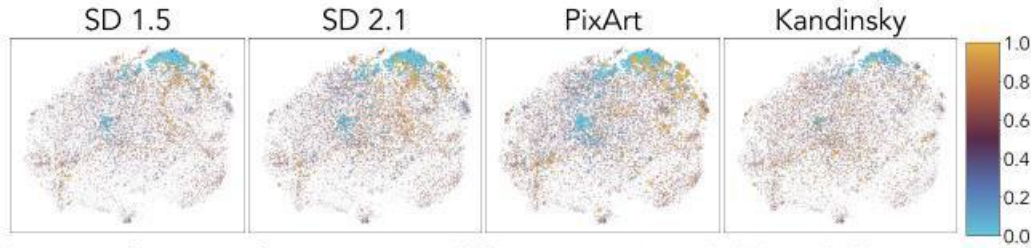


Figure 7: **Structure of concept energy differences.** UMAP embeddings of all 32,000 concepts, colored by their energy difference $\delta(k)$ between the natural and synthesized distributions. Clusters expose structured patterns of conceptual blindspots, with suppressed concepts at the blue end of the color scale and exaggerated concepts at the yellow end.

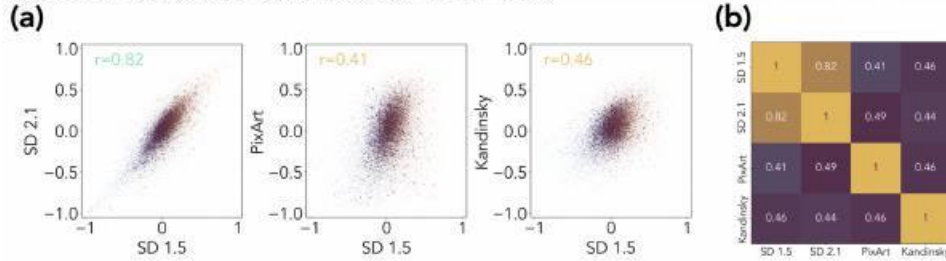


Figure 8: **Cross-model concept energy correlation.** Pairwise scatter plots of $\delta(k)$ across the four evaluated models, with Pearson correlation coefficients annotated top-left. The strong alignment between SD 1.5 and SD 2.1 contrasts with the weaker correlations among the other architectures, evidencing model-specific blindspots. **Correlation matrix of conceptual blindspots.** Heatmap of the pairwise Pearson correlation coefficients of δ across all models, quantifying the degree to which conceptual blindspots are shared.

remaining models. As anticipated, SD 1.5 and SD 2.1 are strongly correlated ($r = 0.82$), reflecting their shared architecture and training pipeline. Their correlations with PixArt and Kandinsky are substantially weaker— $r = 0.41$ and $r = 0.46$, respectively—indicating that these architectures allocate blindspots to different regions of concept space.

In summary, the foregoing analysis indicates that certain blindspots are shared across all models—plausibly a consequence of dataset-level properties—while others are strongly model-specific, arising from idiosyncrasies in training dynamics or model capacity. This dichotomy motivates the separate treatment, in the subsequent sections, of blindspots that are shared across models and those unique to individual models.

5.3 Qualitative Examples of Blindspots

We next present concrete instances of suppressed and exaggerated blindspots to characterize which concepts fall into each regime. Fig. 9a shows a conceptual blindspot suppressed across all models: every evaluated model fails to reproduce the concept `solid white on documents`. Although the conditioning caption explicitly references this concept, none of the generated images realize the intended visual semantics, indicating that this region of concept space is systematically under-sampled across models. Conversely, Fig. 9b illustrates a model-specific blindspot: the concept `pan` is faithfully rendered by three models but conspicuously absent from Kandinsky’s generations. This observation corroborates the finding of Sec. 5.2 that cross-model agreement is high for some concepts and limited for others.

Turning to exaggeration, Fig. 10 presents a case in which the concept `shadow under animal` is over-emphasized in generated images. Although such shadows are individually plausible, their consistent and pronounced rendering across models—relative to the more subtle and variable occurrences in natural images—is symptomatic of an overactive prior. Notably, despite deliberate search, we identified no concept uniquely exaggerated by a single model, suggesting that exaggerations are approximately universal across the evaluated models.

Taken together, these examples concretely demonstrate how conceptual blindspots mani-



fest in generated outputs, and establish that our energy-based diagnostic surfaces both shared and model-specific failure modes. In particular, it exposes non-obvious limitations—such as the consistent failure to reproduce clear or solid background elements, e.g., whitespaces on documents, across all models. This suggests that certain failure patterns may originate from architectural constraints or training-data biases that transcend individual model idiosyncrasies.

While these aggregate-level analyses are informative, they raise a finer-grained question: do blindspots emerge only in aggregate over many samples, or are they detectable at the level of individual datapoints? Answering this permits a more direct interrogation of the underlying mechanisms—distinguishing prompt misinterpretation, latent memorization, or their combination.

5.4 Investigating Datapoint-level Energy Difference: From Incongruent to Memorized Images

To move beyond population-level statistics, we examine individual natural-generated image pairs whose $\delta(\cdot)$ values, averaged over all concepts, attain the largest and smallest magnitudes. This datapoint-level view facilitates interpretation of both successes and failures, the latter of which we find frequently trace to prompt ambiguity or memorization artifacts. Fig. 11a shows pairs with near-zero average $\delta(\cdot)$. In these cases the generated images are conceptually indistinguishable from their natural counterparts. Qualitative inspection, however, indicates that this agreement arises not from faithful capture of the prompt semantics but from verbatim replication of memorized templates: the outputs exhibit repetitive visual structures (e.g., clothing out-

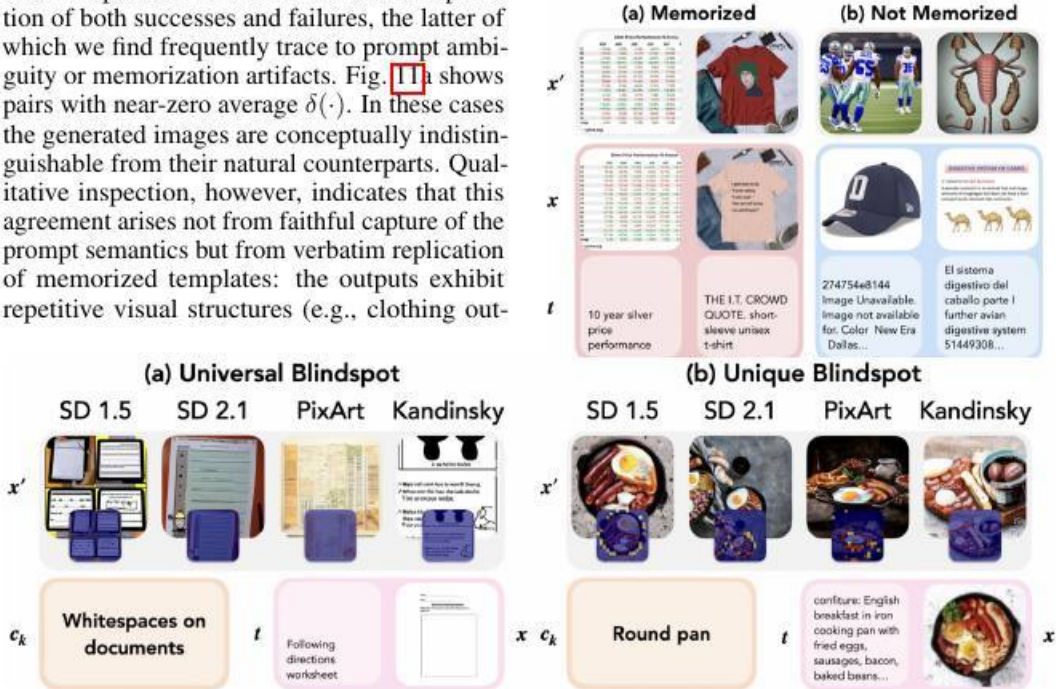


Figure 9: **Examples of suppressed conceptual blindspots.** The natural images x , together with the representative concept c_k and caption t , are shown alongside four synthesized images x' produced by S_θ . The universal blindspot is present across all evaluated models, whereas the unique blindspot is exhibited only by Kandinsky.

387 lines or object arrangements), consistent with
 388 reproduction of over-represented patterns in the
 389 training data. Fig. [11b](#), by contrast, shows pairs
 390 among the largest $\delta(\cdot)$ values. These correspond
 391 to substantial conceptual divergence between the
 392 synthesized and natural image. While some dis-
 393 crepancies are attributable to underspecified or
 394 noisy captions, others reveal genuine blindspots:
 395 the prompt specifies a concept that is faithfully
 396 present in $\mathcal{D}_{\mathcal{X}}$, yet the model fails to realize it
 397 in $\mathcal{D}'_{\mathcal{X}}$. Such failures indicate that, even when language grounding is adequate, certain concepts lie
 398 outside the model’s generative support.

399 5.5 Analyzing Post-Training Effects

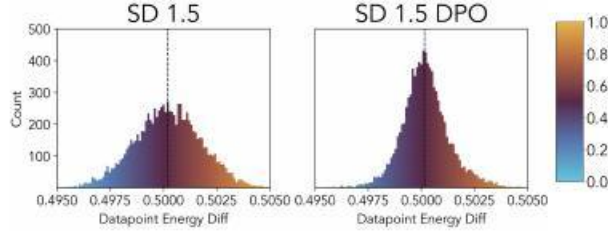


Figure 12: **Effect of DPO on concept fidelity.** Histogram of the datapoint-wise energy difference between the synthesized and natural distributions for the SD 1.5 model with and without DPO.

Post-training protocols such as safety fine-tuning have been argued to reduce the diversity of model generations [140]. Exploiting our pipeline’s ability to isolate discrepancies between a model’s generations and the ground-truth DGP, we apply it to characterize the effects of DPO, a widely used safety fine-tuning protocol [141]. Concretely, we compare two variants of SD 1.5, one trained with DPO and one without. For each image pair $(\mathcal{D}_X, \mathcal{D}'_X)$ we compute the ℓ_2 distance between their concept energy vectors, $\|\xi(\mathcal{D}'_X) - \xi(\mathcal{D}_X)\|_2$. Fig. 12 reports the histogram of these datapoint-wise energy differences. The DPO-tuned model attains both a lower median and a narrower spread, indicating closer distributional agreement between its generated concepts and the ground-truth DGP. This suggests that DPO acts as an implicit regularizer on the model’s concept distribution, biasing outputs toward the semantic content of the observed inputs. While our analysis does not disentangle the specific inductive biases DPO introduces, it provides empirical evidence that the DPO objective—optimizing for human-preferred generations—indirectly improves agreement with the training distribution. Specifically, DPO reduces *both* over- and under-activation of individual concepts relative to the natural baseline. These results underscore the utility of our pipeline for characterizing the downstream effects of post-training interventions—not solely in terms of output quality, but in how such interventions reshape the conceptual geometry of the model’s output space.

416 5.6 Conceptual Misalignment as a Function of Empirical Frequency

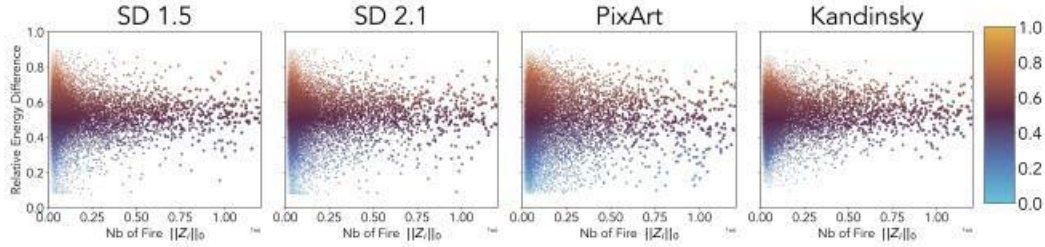


Figure 13: **Concept fidelity across the frequency spectrum.** Scatter plots of the energy difference (y-axis) against concept frequency (x-axis) for the four evaluated models. Each point denotes a concept, with marker size proportional to its activation frequency.

We previously hypothesized that conceptual blindspots are not solely architectural artifacts but may also arise directly from distributional properties of specific features. We test this hypothesis empirically by assessing whether concepts rarely activated in natural images—i.e., those in the long tail of the data distribution—correspond to blindspots in the generative models. Specifically, we pass the natural dataset $\mathcal{D}_X$ through the trained SAE and, for each concept k , compute its empirical frequency $\|Z_{:,k}\|_0$, where $Z_{:,i}$ denotes the activations of concept i across all images. We then relate this frequency to the absolute energy difference measured over the generated outputs. Fig. 13 visualizes this relationship for all evaluated models. High-frequency concepts in natural data tend to exhibit small energy discrepancies, whereas rare concepts—particularly suppressed ones ($\delta(k) < 0.5$)—exhibit substantial alignment errors. This pattern indicates that many blindspots arise not from stochasticity or model-specific quirks but from a systematic dependence on long-tail concept frequency. Mitigating them may therefore require not only architectural modifications but also data-level interventions such as reweighting or augmentation.

6 Discussion

Our analysis identifies numerous conceptual blindspots across four widely used generative image models. The reported results, however, are not exhaustive: each individual finding merits a dedicated investigation in its own right. Rather than pursuing any single finding in depth, we have emphasized the versatility of the proposed method and exploratory tool. Without task-specific modification, they support systematic identification of concepts a model fails to generate, detection of memorization artifacts, discovery of datapoints with insufficient captions, quantification of post-training effects, and characterization of conceptual shifts across architectures. This establishes a foundation for follow-up work to increase the depth of analysis, broaden the set of evaluated architectures, and investigate the root causes of conceptual blindspots. A further direction is the incorporation of hierarchical concept representations within the RA-SAE, which would enable more fine-grained analysis.

Limitations. Several limitations of our work warrant explicit statement. Because concept extraction and representation rely on DINOv2 and the RA-SAE, our analysis is intrinsically confined to the concepts these models encode; concepts poorly represented by either component fall outside its scope. Moreover, although a sample of 10,000 images is substantial, it may under-sample the long tail of rare concepts as well as concept co-occurrence and other compositional statistics.

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Appendix

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A Experimental Setup

This section specifies the experimental setup used to analyze four widely adopted generative image models—SD 1.5/2.1, Kandinsky, and PixArt—each trained on LAION-5B or a subset/derivative thereof. The source code is released in a software repository.

A.1 Observation Space

The observation space is constructed by sampling 10,000 image-text pairs from the LAION-5B dataset [93], which serves as the domain of natural images. Owing to concerns regarding CSAM and other unsafe content, the original release is no longer available; we therefore use a substitute release of a filtered subset, available at <https://huggingface.co/datasets/laion/relaion2B-en-research-safe>

The sampling procedure comprises three steps: (1) loading the full LAION dataset via the Hugging Face datasets library; (2) validating each candidate to confirm well-formed URL structure and resource availability through HTTP HEAD requests; and (3) sampling with replacement until 10,000 valid samples are obtained. This procedure yields D_G as a set of (x, t) tuples, ensuring that the observation space consists of accessible image-text pairs despite the fact that a substantial fraction of the original image URLs have become unavailable since release. Additional synthesized-image examples are provided in Appendix H

A.2 Synthesized Images

For each of the four evaluated models, we generate a synthetic dataset D_{g_θ} in one-to-one correspondence with D_G , producing triplets (x, x', t) . Concretely, for each of the 10,000 image-text pairs (x, t) in D_G , we condition the generative model g_θ on t to synthesize a counterpart image x' .

Synthesis follows the standard text-to-image generation pipeline of each architecture, implemented with the Hugging Face `diffusers` library and with model weights loaded at mixed precision (fp16). All synthetic images are generated at 512×512 resolution using default hyperparameters.

Stable Diffusion 1.5. We use the checkpoint at <https://huggingface.co/benjamin-paine/stable-diffusion-v1-5> (a mirror of the deprecated <https://huggingface.co/runwayml/stable-diffusion-v1-5>). Inference uses 50 denoising steps with the guidance scale fixed at 7.5.

Stable Diffusion 1.5 + DPO. The DPO variant of SD 1.5 (used in the analysis of Sec. 5.5) is identical to the baseline SD 1.5 configuration except that the UNet is replaced with the DPO-trained variant at <https://huggingface.co/mhdang/dpo-sd1.5-text2image-v1>

Stable Diffusion 2.1. We use the checkpoint at <https://huggingface.co/stabilityai/stable-diffusion-2-1>. Inference uses 50 denoising steps with the guidance scale fixed at 7.5.

Kandinsky. We use the checkpoint at <https://huggingface.co/kandinsky-community/kandinsky-2-1>. Inference uses 100 denoising steps with the guidance scale fixed at 4.0.

PixArt. We use the checkpoint at <https://huggingface.co/PixArt-alpha/PixArt-XL-2-1024-MS>. Inference uses 50 denoising steps with the guidance scale fixed at 7.5.

A.3 Distribution Level Analysis

Section 5.1 We compute the energy difference $\delta(\cdot)$ over all 32,000 concepts for each evaluated model. Normalization applies the sigmoid transformation with temperature $T = 0.8$. The resulting values are visualized as log-scale density histograms with 100 bins spanning $[0, 1]$.

Section 5.2 We embed the full set of 32,000 concepts into two dimensions via UMAP applied to the sparse concept codes. Each point corresponds to an individual concept, colored by its energy difference $\delta(\cdot)$ so as to highlight both suppressed and exaggerated blindspots. To quantify cross-model consistency, we compute pairwise Pearson correlation coefficients between the $\delta(\cdot)$ vectors

of all model pairs, yielding both scatter plots and correlation matrices. This analysis determines whether blindspots cluster in concept space and distinguishes model-specific from universal blindspot patterns.

Section 5.3 We rank all 32,000 concepts by their energy difference $\delta(\cdot)$ and manually inspect the extrema (both suppressed and exaggerated blindspots). Suppressed blindspots are selected as concepts with $\delta(\cdot) < 0.1$; exaggerated blindspots as those with $\delta(\cdot) > 0.9$. The presented examples are manually annotated with textual concept descriptions obtained by inspecting their most-activating images and spatial attention patterns. Ongoing efforts to automate this concept interpretation are described in Appendix D

Section 5.5 We compare SD 1.5 with and without DPO as follows. For each image pair (x, x') , we compute the ℓ_2 norm of the difference between their concept energy vectors, $\|\xi(x') - \xi(x)\|_2$. Prior to averaging, a sigmoid transformation with temperature $T = 0.8$ is applied to the element-wise differences. This yields a datapoint-wise energy difference quantifying the deviation of each generated image from its natural counterpart in concept space. The resulting differences are visualized as overlapping histograms contrasting the two model variants.

Section 5.6 For each concept c_k , we compute its empirical frequency $\|Z_{:,i}\|_0$, i.e., the number of non-zero activations across the natural dataset. A sigmoid normalization with temperature $T = 0.4$ is then applied to the energy differences $\delta(\cdot)$. The relationship is visualized as scatter plots in which the x-axis is the empirical concept frequency and the y-axis is the sigmoid-transformed energy difference; marker size is proportional to activation frequency and marker color is proportional to the magnitude of the energy difference.

A.4 • Datapoint Level Analysis

Section 5.4 For each image pair (x, x') , we compute the ℓ_2 norm of the difference between their concept energy vectors, $\|\xi(x') - \xi(x)\|_2$, yielding a scalar measure of conceptual divergence per pair. Pairs are then ranked by this quantity: minimal divergence is indicative of memorization artifacts, whereas maximal divergence signals substantial conceptual failure. This ranking enables targeted qualitative inspection of specific failure modes.

A.5 • Co-occurrence Analysis

Appendix F For both the natural and synthesized datasets D_G and D_{g_θ} , concept co-occurrence structure is analyzed via the co-activation matrix $Z^T Z$, whose entries encode pairwise correlations in concept usage. We characterize the dominant conceptual directions by eigendecomposition of this matrix. Alignment between the natural and synthetic co-occurrence structures is then assessed through cosine-similarity heatmaps between the top-100 eigenvectors of the two matrices. Each resulting 100×100 similarity matrix is rendered as a square heatmap, in which perfect diagonal alignment would indicate identical principal concept axes and off-diagonal structure would reveal rotations and mismatches in compositional geometry.

B Custom RA-SAE Training Details

This section specifies the training configuration of our custom archetypal SAE (RA-SAE) [56]. The trained model is released in a software repository.

Dataset. The autoencoder is trained on the full ImageNet-1k training split (≈ 1.28 M RGB images). Each image is encoded into 261 visual tokens by DINOv2 [135], and these tokens are supplied to the SAE without class or positional embeddings. The total training-token count is therefore $50 \times 1.28\text{M} \times 261 \approx 1.67 \times 10^{10}$.

Dictionary. The dictionary comprises 32,000 concept dimensions. Sparsity is enforced by top- k masking with $k = 5$: for each input, all activations outside the five largest are set to 0. Weights are initialized via Xavier/Glorot. Training is performed at mixed precision (fp16), with the final ten epochs conducted at full precision.

Optimizer and Schedule. The model is trained for 50 epochs with the AdamW optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.999$) and weight decay 10^{-5} . The learning rate follows a linear warm-up over the first 5% of steps, succeeded by cosine decay from $\eta_{\max} = 5 \times 10^{-4}$ to $\eta_{\text{final}} = 10^{-6}$. The objective combines an MSE reconstruction loss with an auxiliary term, weighted by $\lambda = 10^{-5}$, that penalizes activations never entering the top- k set.

972 C Computational Resources

973 This section summarizes the GPU resources consumed in training and experimentation for this paper,
974 totaling approximately 202 GPU-hours on NVIDIA H100 and H200 accelerators.

975 **RA-SAE.** Training required approximately 24 GPU-hours on three NVIDIA H100s.

976 **Synthesized Images.** Generating the complete D_{go} (see Appendix A) required roughly 5 hours per
977 generator when distributed across four NVIDIA H200 GPUs. Aggregated over five generators, this
978 amounts to approximately 100 GPU-hours on the NVIDIA H200.

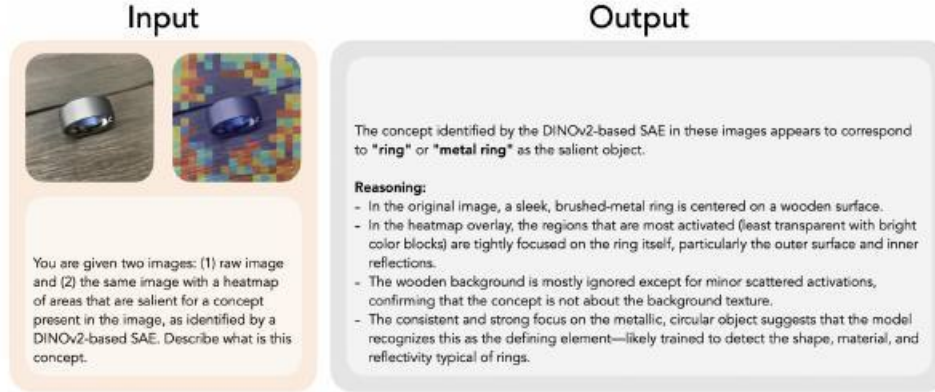
979 **● Distribution-Level Analysis.** Extracting distribution-level energy differences required approxi-
980 mately 3 hours per generator on a single NVIDIA H200 GPU (~ 15 GPU-hours in total).

981 **● Datapoint-Level Analysis.** Computing datapoint-level energy differences, ranking concepts per
982 datapoint, and ranking datapoints per concept likewise required approximately 3 hours per generator
983 on a single H200 GPU (~ 15 GPU-hours in total).

984 D Autointerpretability

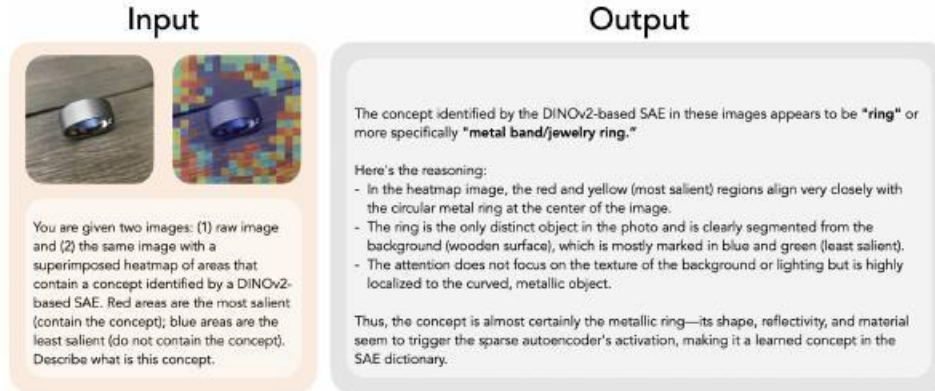
985 To automate the interpretation of concepts recovered by the RA-SAE, we conducted a qualitative
 986 evaluation of the capacity of Vision-Language Models (VLMs) to describe them. Specifically, we
 987 queried ChatGPT 4o through the OpenAI API using a range of prompts and image formats derived
 988 from an image x'' in which the target concept is salient, and recorded the description returned by the
 989 VLM. This section reports the findings of that exploration. For evaluation, we permitted the VLM to
 990 emit its full reasoning trace; a scalable deployment would instead constrain the prompt to elicit the
 991 description alone, suppressing the reasoning.

992 We first supplied the VLM with two images: a raw copy of x'' and a copy of x'' overlaid with a
 993 heatmap localizing the intensity of the concept:



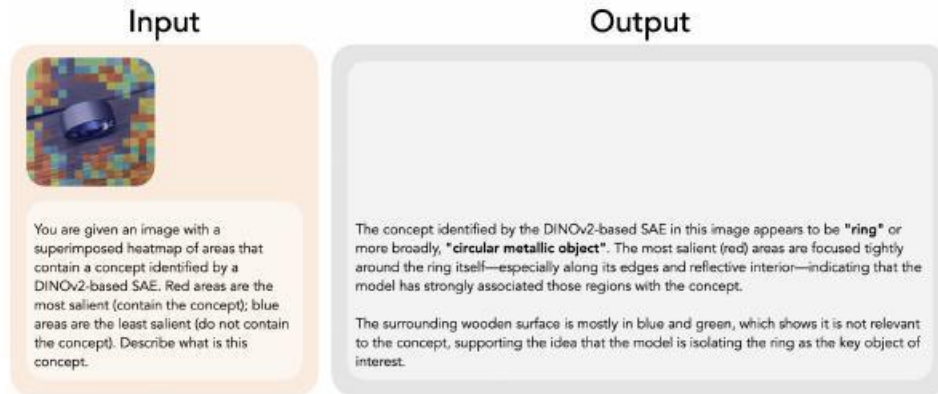
994 As the example shows, the VLM failed to interpret the heatmap correctly: rather than describing
 995 the wood texture in the background—the region actually salient to the RA-SAE—it described the
 996 foreground ring, which the RA-SAE did not attend to.

997 To determine whether this failure stemmed from the prompt, we repeated the experiment with an
 998 expanded prompt providing a more detailed explanation of the heatmap:



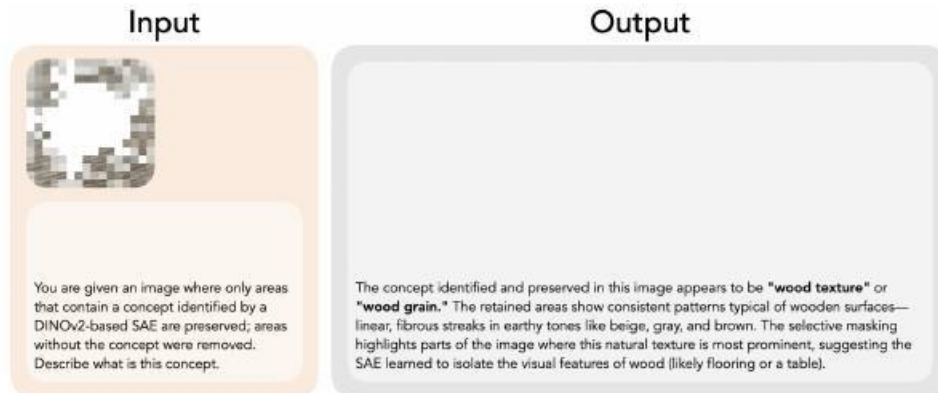
999 This modification did not resolve the issue: the VLM continued to identify the ring as the dominant
 1000 concept. Several additional prompting strategies—including ones describing the heatmap in still
 1001 greater detail, e.g., by supplying the code used to generate it—yielded comparable results.

1002 We next supplied the VLM with only the heatmap-overlaid x'' , together with an analogous prompt
 1003 explaining how to interpret the heatmap:



1004 Again the VLM failed to identify the wooden background as the dominant concept, instead focusing
 1005 on the ring. As before, varying the prompting strategy under this image format left the results largely
 1006 unchanged.

1007 We therefore concluded that the heatmap alone is insufficient to direct the VLM's attention to the
 1008 salient regions, as the presence of less relevant regions appears to distract the model. To mitigate this,
 1009 we replaced the heatmap-overlaid x'' with an alpha-masked version of x'' :



1010 As the example shows, the VLM now correctly identified wood texture as the dominant concept,
 1011 and this success was consistent across a range of concepts and images.

1012 The image formatting of x'' and the prompt described above yielded the most reliable results in our
 1013 qualitative evaluation. We emphasize, however, that this evaluation is limited both by its qualitative
 1014 nature—owing to the absence of ground-truth annotations—and by its restriction to a single VLM.
 1015 We hope that future work on the autointerpretability of SAE concepts can build upon and extend this
 1016 analysis.

1017 E Additional Results: Model-Specific Blindspots



Figure 14: Representative conceptual blindspots identified in **Stable Diffusion 1.5**. For each concept, we display the prototypical image—natural for suppressed blindspots and synthesized for exaggerated blindspots—selected by highest absolute activation, with the concept’s spatial heatmap overlaid.

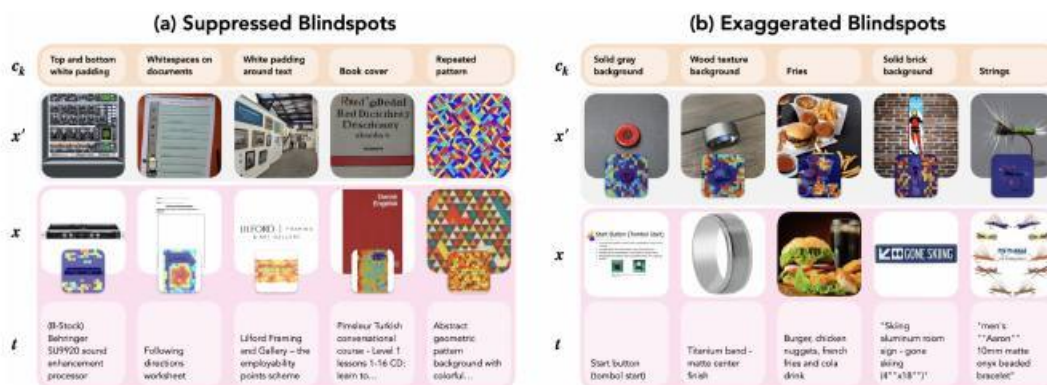


Figure 15: Representative conceptual blindspots identified in **Stable Diffusion 2.1**. For each concept, we display the prototypical image—natural for suppressed blindspots and synthesized for exaggerated blindspots—selected by highest absolute activation, with the concept’s spatial heatmap overlaid.

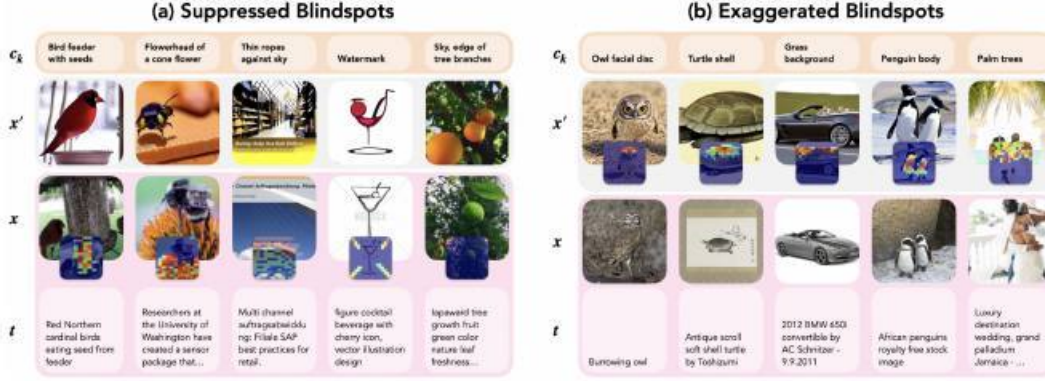


Figure 16: Representative conceptual blindspots identified in **Kandinsky**. For each concept, we display the prototypical image—natural for suppressed blindspots and synthesized for exaggerated blindspots—selected by highest absolute activation, with the concept’s spatial heatmap overlaid.



Figure 17: Representative conceptual blindspots identified in **PixArt**. For each concept, we display the prototypical image—natural for suppressed blindspots and synthesized for exaggerated blindspots—selected by highest absolute activation, with the concept’s spatial heatmap overlaid.

F Additional Results: Higher-order Blindspots with Compositional Discrepancy

Our analysis has thus far centered on individual concept activations. Visual scenes, however, are rarely composed of isolated concepts; they are instead structured by rich co-occurrence patterns that encode compositional semantics. We now assess whether generative models capture this higher-order structure by analyzing the co-activation matrix $Z^T Z$, whose entries encode pairwise correlations in concept usage.

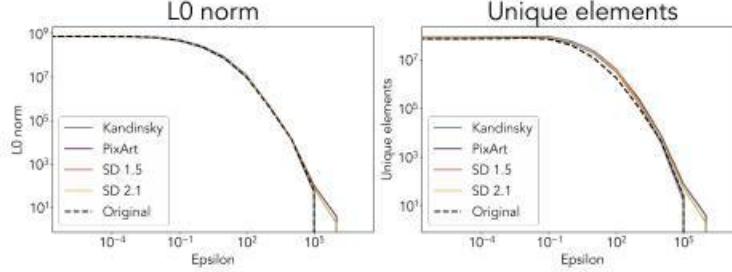


Figure 18: **Sparsity and structural divergence.** Left: the ℓ_0 norm of the co-occurrence matrix ZZ^T as a function of the threshold ϵ , indicating the number of entries that remain active in each model. Right: the number of entries unique to the synthesized distribution relative to the natural distribution. All evaluated models preserve the global sparsity structure while diverging in activation content.

Notably, at the level of binary (thresholded) structure, diffusion models approximate the global sparsity of the natural co-occurrence matrix with high fidelity. As Fig. 18 (left) shows, the ℓ_0 norm of $Z^T Z$, thresholded across a range of ϵ values, tracks closely between the natural and synthesized distributions for all models. This indicates that the coarse connectivity of the conceptual graph—i.e., which concepts co-activate at all—is well preserved. Formally, $Z^T Z$ may be regarded as the adjacency matrix of a weighted, undirected graph over concepts, with edge weights given by co-activation strength across the dataset.

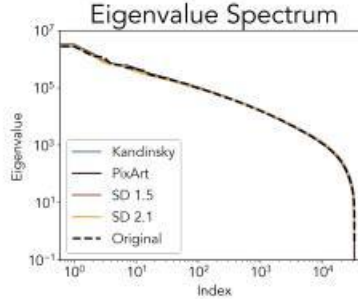


Figure 19: **Spectral structure of co-occurrence.** Log-log plot of the eigenvalue spectra of the co-occurrence matrices ZZ^T for each model and for the natural distribution. All evaluated models reproduce the heavy-tailed decay of the natural distribution.

However, as Fig. 18 (right) illustrates, the specific content of these co-activations diverges: a substantial fraction of the entries in the model-generated $Z^T Z$ are absent from the natural baseline. This indicates that, although the capacity for compositionality is retained, the identity of the active pairings shifts—plausibly reflecting model-specific inductive biases or training artifacts. To probe the internal structure of these co-occurrence patterns, we apply spectral analysis. Fig. 19 reports the eigenvalue spectra of the co-occurrence matrices for each model and for the natural distribution. All spectra exhibit heavy-tailed decay consistent with power-law behavior, indicating that the generative models preserve the overall rank structure and the allocation of variance across conceptual dimensions.

We further examine the alignment of the dominant conceptual directions through cosine-similarity heatmaps between the top-100 eigenvectors of the synthesized and natural co-occurrence matrices

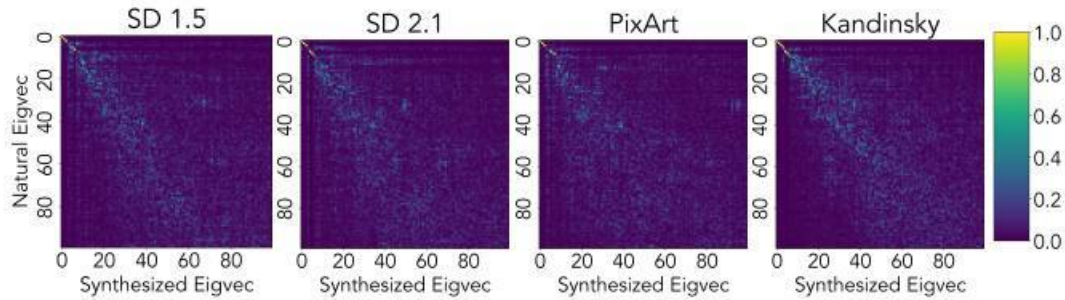


Figure 20: ● **Concept basis similarity.** Cosine-similarity heatmaps between the top 100 eigenvectors of the natural and synthesized co-occurrence matrices ZZ^T . Diagonal structure indicates alignment of the dominant conceptual directions, with the degree of alignment varying across the four models.

1043 (Fig. 20). While all models exhibit partial diagonal alignment—indicating overlap in the princi-
 1044 pal concept axes—the off-diagonal entries reveal rotations and mismatches in the higher modes,
 1045 evidencing systematic deviations in compositional geometry.

1046 Taken together, these findings show that diffusion models approximate the global shape of concept
 1047 co-activation with notable fidelity, yet deviate in subtle, structured ways under spectral analysis.
 1048 Such higher-order discrepancies may underlie failures in generating coherent multi-object scenes or
 1049 relational concepts.

1050 G Additional Results: Blindspot Stress Testing

1051 To stress-test the blindspots identified by our method, we assembled a set of prompts describing each
1052 blindspot, used them to generate a large number of images, and contrasted the outputs of models for
1053 which the concept was flagged as a blindspot against those for which it was not.

1054 Concretely, ChatGPT-4o was prompted as follows: *I want to generate an image of the following*
1055 *concept: "<blindspot>". Suggest 50 prompts highlighting this concept to be used as input for a*
1056 *text-to-image model. Return these as a list of strings in Python.* For each prompt, five images were
1057 generated and scored with our custom RA-SAE (see Appendix B), which ranked them by the intensity
1058 of the target concept. All images were subsequently reviewed manually to determine whether the
1059 blindspot was successfully depicted.

1060 As the following examples of suppressed concepts show, although partial aspects of the target concept
1061 occasionally appeared (e.g., a holder or string for the bird feeder blindspot, and a round hole
1062 for the glossy DVD disc blindspot), the models generally failed to render the full concept. This
1063 outcome is consistent with our method’s assessment and thereby supports the validity of the stress
1064 test.

1065 G.1 Bird Feeder Blindspot in Kandinsky



Figure 21: Images generated from a range of prompts involving the bird feeder concept during stress testing. Our method flagged this concept as a suppressed conceptual blindspot in Kandinsky, consistent with the observed behavior: the model cannot generate a corresponding image. By contrast, SD 1.5—for which the concept was not flagged as a blindspot—generates it successfully.



Figure 22: Images generated from a range of prompts involving the `glossy DVD disc` concept during stress testing. Our method flagged this concept as a suppressed conceptual blindspot in SD 1.5, consistent with the observed behavior: the model cannot generate a corresponding image. By contrast, Kandinsky—for which the concept was not flagged as a blindspot—generates it successfully.

1067 **H Additional Examples: Synthesized Images**



Figure 23: Additional image-caption pairs from LAION-5B, paired with images synthesized from the identical prompt by SD 1.5, SD 2.1, PixArt, and Kandinsky.

1068 I Concentration bounds for δ

1069 In our experiments we estimate $\delta(k)$ from $n = 10,000$ paired samples per concept. As this budget
 1070 is modest, it is natural to ask whether it suffices for reliable estimation. To address this, we derive
 1071 a concentration bound on the empirical estimator $\hat{\delta}_n(k)$ via McDiarmid’s inequality [142]. The
 1072 resulting bound is tight and establishes that accurate estimates of concept bias are attainable even
 1073 from relatively few samples.

1074 **Theorem 4** (Concentration of $\hat{\delta}_n(k)$). *We assume that the concept score $\xi_k(\mathbf{x})$ takes values in $[a, b]$
 1075 almost surely for all images $\mathbf{x}$ drawn from either $\mathcal{D}_X$ or $\mathcal{D}'_X$. Let n paired samples $(\mathbf{x}_i, \mathbf{x}'_i)_{i=1}^n$ be
 1076 drawn independently with $\mathbf{x}_i \sim \mathcal{D}_X$ and $\mathbf{x}'_i \sim \mathcal{D}'_X$, and define the empirical estimator*

$$\hat{\delta}_n(k) := \sigma \left(\frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}_i) - \frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}'_i) \right).$$

1077 Let $M := b - a$ and $L := M/4$. Then for every $\varepsilon > 0$, the deviation satisfies

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right).$$

1078 *Proof.* The function $x \mapsto \sigma(x)$ is $1/4$ -Lipschitz, since $|\sigma'(x)| \leq 1/4$ for all x .

1079 Viewing $\hat{\delta}_n(k)$ as a function of the $2n$ independent variables $(\mathbf{x}_1, \dots, \mathbf{x}_n, \mathbf{x}'_1, \dots, \mathbf{x}'_n)$, changing a
 1080 single argument alters the inner difference of means by at most M/n , and the outer sigmoid scales this
 1081 by at most $1/4$. Hence, the bounded difference constant for each coordinate is $(M/n)(1/4) = L/n$.

1082 By McDiarmid’s inequality [142],

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2\varepsilon^2}{\sum_{j=1}^{2n} (L/n)^2} \right) = 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right),$$

1083 which proves the claim. □

1084 In practice, the concept scores $\delta(k)$ are sparse, the majority concentrating near zero and only a few
 1085 attaining values as large as 10. The concentration bound shows that, even for the largest observed
 1086 biases, a sample size of $n = 10,000$ yields an estimator $\hat{\delta}_n(k)$ that deviates from its true value by at
 1087 most a small ε with high probability. This justifies our sampling strategy and confirms that accurate
 1088 bias measurements are attainable from limited data.

1089 J Monotonicity and Calibration-Free Interpretation of δ

1090 Our objective in analyzing blindspots is to rank concepts by the severity of their generative bias, for
 1091 which we use the score $\delta(k)$. One may ask whether this score introduces distortions relative to more
 1092 direct quantities such as the energy gap or the odds ratio. The following result establishes that $\delta(k)$ is
 1093 a strictly increasing reparameterization of both, and therefore inherits their ordering; consequently,
 1094 no calibration is required when using $\delta(k)$ to rank concepts.

1095 **Theorem 5** (Monotonicity and Calibration of $\delta_{g_\theta \leftrightarrow G}$). *For every concept index k define the energy*
 1096 *gap*

$$\Delta_k = \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}'_X}[\xi_k(\mathbf{x}')] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_X}[\xi_k(\mathbf{x})],$$

1097 *the associated odds ratio $\rho_k = \exp(\Delta_k)$, and the energy-difference score*

$$\delta(k) = \frac{1}{1 + \exp(-\Delta_k)} = \frac{\rho_k}{1 + \rho_k}.$$

1098 *Then $\delta(k)$ is a strictly increasing bijection of both Δ_k and ρ_k , so ranking concepts by any one of*
 1099 *$\delta(k)$, Δ_k , or ρ_k produces exactly the same ordering.*

1100 *Proof.* The logistic sigmoid satisfies $\sigma'(x) = \sigma(x)(1 - \sigma(x)) > 0, \forall x \in \mathbb{R}$; hence σ and therefore
 1101 $\delta(k) = \sigma(\Delta_k)$ grow strictly with Δ_k . Because the exponential map is also strictly increasing and
 1102 bijective $\mathbb{R} \rightarrow (0, \infty)$, setting $\rho_k = \exp(\Delta_k)$ preserves order and gives $\Delta_k = \log \rho_k$. Substituting
 1103 this identity into σ yields $\delta(k) = \sigma(\log \rho_k) = \rho_k / (1 + \rho_k)$, which is the composition of two
 1104 strictly increasing bijections and is therefore itself strictly increasing and bijective in ρ_k . Since strict
 1105 monotonic functions never reverse inequalities, the three quantities share the same total order over
 1106 concepts.

1107

□

1108 Consequently, ranking concepts by $\delta(\cdot)$ is strictly equivalent to ranking them by the energy gap or
 1109 by the conceptual generation odds ρ_k . No calibration is necessary, as all three quantities induce the
 1110 identical total order over concepts.

1111 K Stability of FID Under SAE Embeddings

1112 In this section we establish a quantitative relationship between the Fréchet Inception Distance (FID)
 1113 computed in the original activation space (of dimension d) and the FID computed after applying a
 1114 (potentially overcomplete) SAE dictionary $D \in \mathbb{R}^{k \times d}$ with $k \gg d$. Throughout, we assume that D
 1115 has orthonormal columns but is not necessarily square, i.e.,

$$D^\top D = I_d, \quad \text{while} \quad DD^\top \neq I_k.$$

1116 We first recall an elementary fact: when D is square and orthogonal ($k = d$), it acts as an isometry
 1117 between A and Z , so that FID is exactly preserved. As this case is unrealistic, we subsequently
 1118 address the overcomplete case and show that FID can be bounded in terms of the extremal singular
 1119 values of D . We work with the Wasserstein-2 metric $\mathcal{W}_2$, noting that FID coincides with $\mathcal{W}_2^2$
 1120 specialized to Gaussians.

1121 For a probability measure μ on $\mathbb{R}^d$ we denote by $D_\# \mu$ its push-forward under D , i.e., $D_\# \mu(z) =$
 1122 $\mu(D^{-1}z)$. Let $\sigma_{\min}$ and $\sigma_{\max}$ denote the minimal and maximal singular values of D , equivalently
 1123 the square roots of the extremal eigenvalues of $DD^\top$:

$$\sigma_{\min}^2 I_k \preceq DD^\top \preceq \sigma_{\max}^2 I_k.$$

1124 Empirically one typically finds $\sigma_{\min}, \sigma_{\max} \approx 1$, though the proof does not rely on this. We begin
 1125 with a simple lemma for the non-overcomplete case.

1126 **Lemma 1** (Isometry under exact orthogonality). *Suppose $k = d$ and $D^\top D = DD^\top = I_d$. Then*
 1127 *D is an isometry: $\|Dv\|_2 = \|v\|_2$ for all $v \in \mathbb{R}^d$. Consequently, for any probability measures μ, ν*
 1128 *on $\mathbb{R}^d$ with finite second moment,*

$$\mathcal{W}_2(D_\# \mu, D_\# \nu) = \mathcal{W}_2(\mu, \nu).$$

1129 *Proof.* Orthogonality of D preserves the Euclidean norm, and the push-forward commutes with the
 1130 map inside the $\mathcal{W}_2$ infimum; the integrand is therefore unchanged, so the infimum is identical. $\square$

1131 This case, however, is unrealistic, as SAEs typically rely on overcompleteness to extract meaningful,
 1132 interpretable concepts. In the overcomplete case D is no longer orthonormal, but it can still be
 1133 column-orthonormal. We exploit this property to derive a bound in terms of the extremal singular
 1134 values of $D^\top D$.

1135 **Theorem 6** (FID under column orthogonal embeddings). *Let $D \in \mathbb{R}^{k \times d}$ satisfy $D^\top D = I_d$ and*
 1136 *denote by $0 < \sigma_{\min} \leq \sigma_{\max}$ the extreme singular values of $DD^\top$. Given two data matrices*
 1137 *$A, A' \in \mathbb{R}^{n \times d}$ (rows are sample vectors), set $Z = AD^\top \in \mathbb{R}^{n \times k}$ and $Z' = A'D^\top \in \mathbb{R}^{n \times k}$. Then*

$$\sigma_{\min}^2 \text{FID}(A, A') \leq \text{FID}(Z, Z') \leq \sigma_{\max}^2 \text{FID}(A, A').$$

1138 *Proof.* Write μ for the empirical measure of A and ν for that of A' , i.e.

$$\mu = \frac{1}{n} \sum_{i=1}^n \delta_{A_{i,:}}, \quad \nu = \frac{1}{n} \sum_{i=1}^n \delta_{A'_{i,:}}.$$

1139 For any coupling $\pi \in \Pi(\mu, \nu)$ (i.e. a probability measure on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ, ν) we have,
 1140 by the extremal singular value bound,

$$\sigma_{\min}^2 \|x - y\|_2^2 \leq \|D(x - y)\|_2^2 \leq \sigma_{\max}^2 \|x - y\|_2^2, \quad \forall (x, y) \in \mathbb{R}^d \times \mathbb{R}^d.$$

1141 Integrating with respect to an arbitrary coupling $\pi \in \Pi(\mu, \nu)$ yields

$$\sigma_{\min}^2 \int \|x - y\|_2^2 d\pi \leq \int \|D(x - y)\|_2^2 d\pi \leq \sigma_{\max}^2 \int \|x - y\|_2^2 d\pi.$$

1142 The middle integral is exactly the transport cost of the pushed-forward coupling $(D \times D)_\# \pi$ between
 1143 $\mu_D := D_\# \mu$ and $\nu_D := D_\# \nu$. Because the inequalities hold for every π , they hold in particular
 1144 for the optimal couplings attaining $\mathcal{W}_2(\mu, \nu)$ and $\mathcal{W}_2(\mu_D, \nu_D)$, though these two optima need not

1145 coincide. Taking the infimum over π term-wise makes this explicit:

$$\sigma_{\min}^2 \inf_{\pi \in \Pi(\mu, \nu)} \int \|x - y\|_2^2 d\pi \leq \inf_{\pi \in \Pi(\mu, \nu)} \int \|D(x - y)\|_2^2 d\pi \leq \sigma_{\max}^2 \inf_{\pi \in \Pi(\mu, \nu)} \int \|x - y\|_2^2 d\pi.$$

1146 Hence

$$\sigma_{\min}^2 \mathcal{W}_2^2(\mu, \nu) \leq \mathcal{W}_2^2(\mu_D, \nu_D) \leq \sigma_{\max}^2 \mathcal{W}_2^2(\mu, \nu).$$

1147 Recognising $\text{FID}(\cdot, \cdot) = \mathcal{W}_2^2(\cdot, \cdot)$ for the Gaussian surrogate and plugging in (A, A') (resp. (Z, Z'))
 1148 finishes the proof. $\square$

1149 In essence, Theorem 6 states that applying a column-orthogonal overcomplete SAE dictionary distorts
 1150 the Fréchet Inception Distance by at most the square of its extremal singular values. When D is
 1151 nearly orthogonal—empirically, $\sigma_{\min}, \sigma_{\max} \approx 1$ —the result implies that FID measured in the SAE
 1152 feature space is essentially equal to the canonical FID. $\square$

1153